

Roll No.....

END TERM EXAMINATION, JUNE 2024
COURSES: MCA, MSc DATA SCIENCE 2nd SEMESTER
MSc PHYSICS 4TH SEMESTER
SUBJECT: CAREER SKILLS

TIME: 3 Hours

Maximum Marks: 100

Instructions for candidates:

- (i) All questions are compulsory.
- (ii) Each question carries 1 mark.
- (iii) Use of calculator is not allowed.
- (iv) There is no negative marking for incorrect response.

Direction (1-7) Read the following passage given below and answer the questions accordingly:

Between September 2017 and April 2018, as per a CSO report, the economy added 4.1 million new jobs in the formal sector. The CSO release defines jobs as ones that provide at least one government financed (or mandated) social security benefit such as Employees' Provident Fund (EPF), National Pension Scheme, or Employees' State Insurance Scheme. NITI Aayog and official economists have also put out similar estimates since early this year, inviting widespread skepticism among knowledgeable people.

How reliable or credible are the official estimates? Very little and for two reasons: the estimates are based on administrative records of implementing the social security schemes, whose completeness, consistency and accuracy are unknown; and since a formal sector worker, in principle, can legitimately access more than one social security scheme, double counting is a distinct possibility. The release does not explain how the problem is addressed in the database. Therefore, rightly, experts have demanded the release of the administrative data for independent verification.

Moreover, the official data suffer from a conceptual problem. The social schemes mentioned above are applicable to establishments above a certain size (of employment), and to certain kinds of enterprises. For instance, in the factory sector, those employing 20 or more workers are mandated to provide EPF to all the workers. So, if in a factory, employment goes up from 19 to 20 workers, it comes under the purview of the EPF, to be provided to all the 20 workers. Thus, the EPF enrolment increases by 20 workers, but the additional job created is just for one worker. Herein lies the fault.

The formal sector stands at the apex of India's labour market pyramid, agriculture being at the bottom, employing 50% of the workforce. The remaining workers are in the non-farm informal sector, spread across rural and urban areas. In fact, it is this sector that has grown in recent decades at the expense of the other two sectors mentioned above. Moreover, nearly half of the informal labour workers are self-employed in household (or own account) enterprises, often engaging in unpaid family labour. Varying degrees of under-employment or disguised unemployment are the defining feature of informal labour markets.

Since 1972-73, the five-yearly Employment-Unemployment Surveys (EUS) conducted by the National Sample Survey (NSS) have been the mainstay for analyzing labour market trends. Though infrequent, the database has served a valuable purpose of capturing the complexities of the labour market; access to household-level data lately has spawned rich and granular analyses of the informal economy. As the last round of the EUS was held in 2011-12, there is no reliable way of updating employment trends. The EUS has been replaced with an annual Period Labour Force Survey, and a time use survey. However, it will be a while before these data series come up with stable and credible estimates. The Labour Bureau under the Ministry of Labour and Employment has been carrying out household surveys somewhat similar to the EUS since 2010-11. They show a decline in worker-population ratio between 2013-14 and 2015-16, suggesting a deteriorating employment situation recently.

Q1. Which of the following is/are synonyms of skepticism?

- | | | | |
|-------------|-------------------|-------------------|-----------------------|
| I. Cynicism | II. Belief | III. Distrust | IV. Conviction |
| a. only II | b. only II and IV | c. only I and III | d. only I, II and III |

Q2. Which of the following is/are antonyms of spawned?

- | | | | |
|--------------|--------------|-------------------|-----------------------|
| I. Destroyed | II. Engender | III. Procreated | IV. Sired |
| a. only I | b. only IV | c. only II and IV | d. only I, III and IV |

Q3. As per paragraph 1, which of the following could weaken the importance of having added 4.1 million jobs in the economy as per the CSO report?

- I. The report is silent about the majority of the workforce engaged in the informal sector.
- II. The formal sector accounts for just about 12-15% of India's workforce.
- III. The estimates supplement the data on organized sector employment reported regularly by the Ministry of Labour and Employment.

- | | | | |
|-----------|------------|------------------|--------------------|
| a. only I | b. only II | c. only I and II | d. only II and III |
|-----------|------------|------------------|--------------------|

Q4. As per paragraph 1, which of the following could weaken the importance of having added 4.1 million jobs in the economy as per the CSO report?

- I. The report is silent about the majority of the workforce engaged in the informal sector.
- II. The formal sector accounts for just about 12-15% of India's workforce.
- III. The estimates supplement the data on organized sector employment reported regularly by the Ministry of Labour and Employment.

- | | | | |
|-------------|------------------|--------------------|-------------------|
| a. only III | b. only I and II | c. only II and III | d. only I and III |
|-------------|------------------|--------------------|-------------------|

Q5. Which of the following is a suitable gist of paragraph 3?

- a. The social security databases are lists of workers enrolled in the schemes, as an entitlement or as voluntary subscribers and not employment registers.
- b. The social security databases list out workers and accurately informs the actual level of employment in the firm.
- c. The employees have the option to get themselves enrolled in social security schemes on reaching a particular number as specified by the Central and State laws.
- d. The official data is matched with the register for social schemes to avoid any discrepancies in the calculation of the employment figures.

Q6. As per paragraph 4, which of the following is/are some features of the non-farm informal labour markets?

- I. It has grown the most in recent decades.
- II. Almost all of the workers are self-employed or engaged in unpaid labour.
- III. This segment suffers from disguised employment.

- | | | | |
|------------|-------------|------------------|-------------------|
| a. only II | b. only III | c. only I and II | d. only I and III |
|------------|-------------|------------------|-------------------|

Q7. What could be a/some possible reason/s for so few jobs to be created, given that the growth rate is more than 7%?

- I. There is no proper record of job creation in the informal economy, which forms the bulk of employment, and hence no way of accurately knowing how many jobs have been created.
- II. The growth has mostly been in areas which are capital intensive and not labour intensive.
- III. There is a possibility that the GDP figures are overestimated on account of some changes in the new National Accounts Statistics (NAS) series and the economic growth is overstated.

- | | | | |
|------------|-------------|--------------------|---------------------|
| a. only II | b. only III | c. only II and III | d. all of the above |
|------------|-------------|--------------------|---------------------|

Direction (8-13) In the questions given below, rearrange the parts, which are labelled A, B, C, D and E to produce a logical and coherent sequence of sentences.

Q8. A. In simpler terms, it is the Indian version of the Razzies.

B. The 3rd Golden Kela Awards will be hosted by Cyrus Broacha this year.

C. It was created in order to ridicule the bad performances and as a revenge for wasting our precious time and money on such idiotic films.

D. The Golden Kela is held each year where awards are given for the year's worst in Bollywood.

- E. It was created by Random magazine, India's longest running humor magazine in the year 2009.
 a. DAECB b. BACDE c. ACEDB d. CEADB
- Q9. A. Despite the strong performance of the economy in 2010-11, the outlook for 2011-12 is clouded by stubborn and persistently high inflation, and rising external risks.
 B. The three key macroeconomic concerns before the Union Budget 2011-12 were high inflation, high current account deficit (CAD), and fiscal consolidation.
 C. Additionally, there was an expectation that the government would restart the reform process.
 D. While the Budget sets a lower nominal gross domestic product (GDP) growth target of 14%, we believe that the real GDP growth target of 9% factored in the Budget is on the optimistic side.
 E. The Budget has made an attempt to address all these issues, albeit through small steps.
 a. BCEAD b. CBAED c. DACEB d. ADCEB
- Q10. A. These were mainly bulwarks against winter, the hoarded dregs of more plentiful seasons.
 B. The first were the earliest mince pies, which saw cooked, shredded meat, dried fruits, alcohol with its preservative qualities and perhaps a few spices or herbs, all encased in large pies.
 C. Subsequently, people baked this into a kind of pie, adding the breadcrumbs bulk, eggs to bind it, and upping the dried fruits and called it 'plum pudding'.
 D. The pudding seems to have had two principal forerunners.
 E. The second main pudding was a pottage or soup called frumenty, a fast dish involving cracked wheat, currants and almonds which was ladled out at the start of a meal.
 a. ECDAB b. BAECD c. DACEB d. DBAEC
- Q11. A. In a bid to placate the associate members, the ICC has decided to increase the number of participating teams to 16 in the Twenty20 World Cup, as the game's governing body feels these countries will have a greater chance of competing on an equal footing in cricket's shortest format.
 B. It is convenient just now to forget that in the last edition of the tournament, considerable criticism was heaped on the governing body for the inordinate length of the tournament, thanks in large part to the presence of the associates.
 C. To be fair to the ICC, criticism of the move to restrict the number of teams in the next edition of the Cup is a case of damned if you do, damned if you don't.
 D. The ICC's decision to restrict the number of teams in the 2015 World Cup has evoked mixed responses, with opinion divided among players of the full member teams.
 E. Not surprisingly, the associate members aren't too thrilled about the idea of being kept out of cricket's showpiece event.
 a. ECDAB b. BAECD c. DACEB d. DEACB
- Q12. A. Environment Education unit of Centre for Science & Environment has always been working towards providing easy to understand reading material.
 B. Their new publication on this subject is an attempt to lend teachers a helping hand.
 C. It unfolds in two sections: Climate change: how to make sense of it all.
 D. And natural resources how to share & care.
 E. However, they are introduced to students not as a paragraph to memorize but as an activity to do.
 a. ACEBD b. DBCAE c. ABCDE d. BECAD
- Q13. A. A famous Japanese rock garden is at Ryoan-Ji in Northwest Kyoto, Japan.
 B. The rocks of various sizes are arranged on small white pebbles in five groups, each comprising five, two, three, two, & three rocks.
 C. The garden is 30 meters long from East to West & 10 meters from north to south.
 D. The garden contains 15 rocks arranged on the surface of white pebbles in such a manner that visitors can see only 14 of them at once from whichever angle the garden is viewed.
 E. There are no trees, just 15 irregularly shaped rocks of varying sizes, some arranged by gravel/sand that is raked every day.
 a. ACEBD b. CAEDB c. DEABC d. BADEC
- Direction (14-22): In each question given below, read the statements thoroughly and select the best answer.

Q14. There is a piece of folk wisdom expressed in the saying, "If it is not broken, don't fix it." A factory manager who accepted that saying would be least likely to:

- a. Agree to union demands, in the interest of safety, for better lighting in the stairwells and storage areas.
- b. Respond to the difficulty of retaining skilled electronic technicians by establishing an on-site day-care center for small children.
- c. Order the immediate replacement of windows broken in a strike.
- d. Replace the quality control supervisor after receiving several complaints about defective units in recent shipments from the factory.

Q15. Despite recent rumors of a new and improved building, employees **should not expect renovations**. Without the support of the building's supervisor, the committee's plan usually fails. Two years ago, **a plan to renovate the meeting rooms went under after the supervisor changed his mind and withdrew his support**.

The bold phrases play which of the following roles in the argument above?

- a. The first phrase offers advice and the second phrase states a conclusion.
- b. The first phrase states the conclusion and the second phrase supports that conclusion with an analogy.
- c. The first phrase states a conclusion and the second phrase provides evidence that weakens the conclusion.
- d. The first phrase states a premise on which the conclusion is based and the second phrase states the conclusion.

Q16. Policies in the United States encourage high debt. A tax deduction on mortgage interest encourages overspending on housing and taxing capital gainst discourages investments. Nevertheless, U.S. policy isn't doesn't entirely encourage borrowing over savings because it has _____.

Which of the following best completes the idea above?

- a. deductions for workers to reduce their annual tax burden
- b. cut taxes on lower income workers
- c. tax-free retirement savings programs
- d. Federal Bank monetary policy to fight inflation and keep interest rates low.

Q17. Recent research in neuropsychology suggests that laughter is less a response to humor than a kind of involuntary social signal. In one study, a social scientist observed instances of laughter in diverse settings, and found that 80 to 90 percent occurred after straight lines such as "talk to you later" and "I know".

The argument above depends on which of the following unstated assumptions?

- a. Involuntary social signals can be studied with observational methods.
- b. The diverse settings in the study included diverse populations as well.
- c. Statements can be removed from context without any loss of meaning.
- d. People laugh after straight lines because they believe they are expected to.

Q18. Streptococcus mutans, the bacteria that causes tooth decay, is transmitted almost exclusively from mother to child, which allows it to be traced back thousands of years via its DNA to one of several distinct lineages. This would not be possible if it were transmitted more easily from person to person.

Which of the following can be inferred from the passage above?

- a. Mothers can be blamed for tooth decay in children.
- b. Streptococcus mutans can be classified according to lineage.
- c. If it were transmitted from father to child, streptococcus mutans would not be traceable.
- d. Bacteria cannot be traced if it has no DNA.

Q19. After studying ceramic pieces found in a seaport in Israel, archaeologists determined that the Philistines, an ancient people who settled in the Aegean Sea region around 1200 BC; were able to read and write in an undeciphered, non-Semitic language. This proves that, contrary to previous beliefs, the Philistines were actually a cultured and literate people.

Which of the following, if true, offers the most support for the conclusion drawn in the passage above?

- a. Tamil, a classical language of India and Sri Lanka, dates back to 500 BC.
- b. The Philistines also had refined pottery and sophisticated architecture.
- c. Archaeological research suggests that the Philistines had cultural ties to the Myceneans, in Greece.
- d. The Philistines held a monopoly on iron smithing consistent with the Biblical descriptions of Goliath's armor.

Q20. In the secretarial schools of the mid-20th century, accuracy of typing was valued as much as speed. Business letters were supposed to be error-free with no visible corrections, and on typing tests, deductions were made from the

speed scoring for every error found. The development of so many varieties of correction methods and technologies in this time period is therefore somewhat of a mystery.

Which of the following, if true, would most weaken the conclusion drawn above?

- a. Business letters were typed on a heavyweight paper that not only looked good, but could endure repeated erasure.
- b. A special kind of typewriter paper called erasable bond was invented to allow easier removal of type from the page.
- c. Typewriters are more difficult to type on than keyboards, and even the most sophisticated typewriters would sometimes jam.
- d. In the mid-20th century, typewriters were as ubiquitous as computers are today.

Q21. Felicia: Internships are a waste of time and energy. In the time you spend working for free, you could be temping or taking classes or pursuing employment that not only fulfills your interests, but also compensates you for your hard work.

Nitesh: The purpose of internships isn't compensation, it's preparation for a future career. With an internship, you can secure connections within your field of choice and also gain experience that looks great on a resume.

Nitesh objects to Felicia's argument by:

- a. pointing out an error of logic.
- b. refuting a previously agreed-upon conclusion.
- c. objecting to an unstated premise.
- d. suggesting an alternative definition.

Q22. Although it's a common perception that philosophy has no practical application to the real world, **history suggests otherwise**. In fact, the word "philosophy" itself derives from the Greek roots philos, meaning "love", and sophos, which means "wisdom". Taken together, **the word "philosophy" literally means "love of wisdom."**

The bold phrases play which of the following roles in the argument above?

- a. The first phrase states the conclusion and the second phrase offers support for that conclusion.
- b. The first phrase introduces evidence supporting a conclusion, and the second phrase contains that evidence.
- c. The first phrase contains an objection to a common perception, and the second phrase offers support for that objection.
- d. The first phrase states a premise on which the conclusion is based, and the second phrase offers a supporting definition.

Directions (23-28): Replace the underlined portion with one of the given options, which results in a sentence that is clear, precise, and meets the requirements of standard written English.

Q23. In addition to helping the environment, renewable resources—such as solar power and hydroelectricity—may also provide an economic benefit, as it may reduce our dependence on foreign oil.

- a. as they may reduce our dependence on foreign oil.
- b. as they depend on the reduction of foreign oil.
- c. as it may reduce our dependence on foreign oil.
- d. as it reduces our dependence on foreign oil.

Q24. Most men who go to sporting events find it an enjoyable diversion.

- a. Most men who go to sporting events find them an enjoyable diversion.
- b. Most men who go to sporting events find it to be an enjoyable diversion.
- c. Many men who go to sporting events find it an enjoyable diversion.
- d. Most of the men who go to sporting events find it an enjoyable diversion.

Q25. While it can seem like a daunting task to train for a marathon, anyone who has completed the challenge knows that preparing a plan that they stick to is the key to success.

- a. sticking to a plan
- b. preparing a plan that they stick to
- c. preparing a plan that he sticks to
- d. preparing a plan and sticking to it

Q26. Just as they finished work for the day, the worker found his lost green helmet and made his way home happily.

- a. Just as they finished work for the day, the worker found his lost green helmet and made his way home happily.
- b. Just as he finished work for the day, his lost green helmet was found by the worker, who made his way home happily.
- c. Just as he finished work for the day, the worker found his lost green helmet and made his way home happily.

- d. Just as they finished work for the day, the worker finds his lost green helmet and made his way home happily.

Q27. Although the basketball team used their star players, the match was lost.

- a. Although the basketball team used their star players
- b. Although the star players were used by the basketball team
- c. Although the basketball team used its star players
- d. The basketball team used its star players, although

Q28. A student should always have access to their notes.

- a. one's notes
- b. its notes
- c. his or her notes
- d. it's notes

Directions (29-36) In the sentences given below, select one option as your answer out of the four so that the sentences follow parallel structure.

Q29. When Sema realized that her father had made lima beans and rice for dinner, she _____, claimed to feel nauseous, and excused herself from the table. The bag of stale popcorn in her room would tide her over until breakfast.

- a. clutched her stomach
- b. did clutch her stomach
- c. was clutching her stomach
- d. none of the above

Q30. Not only did Harshita despise the way June chewed with her mouth open, _____ finding her wet towels all over the bathroom floor.

- a. but also disliking
- b. but also disliked
- c. but he also disliked
- d. none of the above

Q31. To protect her delicate hands, Farah will not rake the yard; do the dishes, or _____.

- a. scrub the bathtub
- b. won't scrub the bathtub
- c. refuse to scrub the bathtub
- d. none of the above

Q32. When Disha takes her beagle for a walk, Santana enjoys sniffing for edible garbage, _____, and lunging at squirrels.

- a. she howls at bicyclists
- b. howls at bicyclists
- c. howling at bicyclists
- d. none of the above

Q33. After Amrita cashes her paycheck, the money goes to her savings account, cappuccino fund, and _____.

- a. credit card balance
- b. paying her credit card bill
- c. towards her credit card balance
- d. none of the above

Q34. In preparation for her run, Anshika _____, applied sunscreen, and increased the volume on her iPod.

- a. tightened her shoelaces
- b. was tightening her shoelaces
- c. did tighten her shoelaces
- d. none of the above

Q35. We searched the car trunk, _____, and the top of the refrigerator, but we could not find the box of cornflakes that we remember buying.

- a. on the pantry shelves
- b. the pantry shelves
- c. looked on the pantry shelves
- d. none of the above

Q36. Bonku, our ninety-pound golden retriever, chews on furniture, _____, and snots up the car windows—Inconveniences we didn't anticipate when we adopted him.

- a. would drag us around the neighborhood during walks.
- b. drags us around the neighborhood during walks.
- c. he drags us around the neighborhood during walks.
- d. none of the above

Direction (37-40) Read the following passage below and answer the questions accordingly:

The first systems of writing developed and used by the Germanic peoples were runic alphabets. The runes functioned as letters, but they were much more than just letters in the sense in which we today understand the term. Each rune was an ideographic or pictographic symbol of some cosmological principle or power, and to write a rune was to invoke and direct the force for which it stood. Indeed, in every Germanic language, the word "rune" (from Proto-Germanic *runo) means both "letter" and "secret" or "mystery," and its original meaning, which likely predated the adoption of the runic alphabet,

may have been simply "(hushed) message."

Each rune had a name that hinted at the philosophical and magical significance of its visual form and the sound for which it stands; which was almost always the first sound of the rune's name. For example, the T-rune, called *Tiwaz in the Proto-Germanic language, is named after the god Tiwaz (known as Tyr in the Viking Age). Tiwaz was perceived to dwell within the daytime sky, and, accordingly, the visual form of the T-rune is an arrow pointed upward (which surely also hints at the god's martial role). The T-rune was often carved as a standalone ideograph, apart from the writing of any particular word, as part of spells cast to ensure victory in battle.

The runic alphabets are called "futharks" after the first six runes (Fehu, Uruz, Thurisaz, Ansuz, Raidho, Kaunan), in much the same way that the word "alphabet" comes from the names of the first two Hebrew letters (Aleph, Beth). There are three principal futharks: the 24-character Elder Futhark, the first fully-formed runic alphabet, whose development had begun by the first century CE and had been completed before the year 400; the 16-character Younger Futhark, which began to diverge from the Elder Futhark around the beginning of the Viking Age (c. 750 CE) and eventually replaced that older alphabet in Scandinavia; and the 33-character Anglo-Saxon Futhorc, which gradually altered and added to the Elder Futhark in England. On some inscriptions, the twenty-four runes of the Elder Futhark were divided into three ættir (Old Norse, "families") of eight runs each, but the significance of this division is unfortunately unknown.

Runes were traditionally carved onto stone, wood, bone, metal, or some similarly hard surface rather than drawn with ink and pen on parchment. This explains their sharp, angular form, which was well-suited to the medium. Much of our current knowledge of the meanings the ancient Germanic peoples attributed to the runes comes from the three "Rune Poems," documents from Iceland, Norway, and England that provide a short stanza about each rune in their respective futharks (the Younger Futhark is treated in the Icelandic and Norwegian Rune Poems, while the Anglo-Saxon Futhorc is discussed in the Old English Rune Poem).

While runologists argue over many of the details of the historical origins of runic writing, there is widespread agreement on a general outline. The runes are presumed to have been derived from one of the many Old Italic alphabets in use among the Mediterranean peoples of the first century CE, who lived to the south of the Germanic tribes. Earlier Germanic sacred symbols, such as those preserved in northern European petroglyphs, were also likely influential in the development of the script.

The earliest possibly runic inscription is found on the Meldorf brooch, which was manufactured in the north of modern-day Germany around 50 CE. The inscription is highly ambiguous, however, and scholars are divided over whether its letters are runic or Roman. The earliest unambiguous runic inscriptions are found on the Vimose comb from Vimose, Denmark and the Øvre Stabu spearhead from southern Norway, both of which date to approximately 160 CE. The earliest known carving of the entire futhark, in order, is that on the Kylver stone from Gotland, Sweden, which dates to roughly 400 CE. The transmission of writing from southern Europe to northern Europe likely took place via Germanic warbands, the dominant northern European military institution of the period, who would have encountered Italic writing firsthand during campaigns amongst their southerly neighbors. This hypothesis is supported by the association that runes have always had with the god Odin, who, in the Proto-Germanic period, under his original name *Wōðanaz, was the divine model of the human warband leader and the invisible patron of the warband's activities. The Roman historian Tacitus tells us that Odin ("Mercury" in the interpretatio romana) was already established as the dominant god in the pantheons of many of the Germanic tribes by the first century.

From the perspective of the ancient Germanic peoples themselves, however, the runes came from no source as mundane as an Old Italic alphabet. The runes were never "invented," but are instead eternal, pre-existent forces that Odin himself discovered by undergoing a tremendous ordeal.

Q37. The word "pantheon" in the passage refers to

- a. A temple of all the gods
- b. All the gods collectively of a religion

- c. A monument or building commemorating a nation's dead heroes
d. A domed circular temple at Rome, erected a.d. 120–124 by Hadrian
- Q38. Which of the following statements is incorrect?
- a. Unlike the Latin alphabet, which is an essentially utilitarian script, the runes are symbols of some of the most powerful forces in the cosmos
b. Runic writing was probably first used in southern Europe and was carried north by Germanic tribes.
c. The word "rune" and its meaning was derived from the runic alphabet.
d. The first runic alphabets date back to the 1st century CE.
- Q39. Which of the following can be inferred from the passage?
- I. Runic script was most likely derived from Old Italic script.
II. Runes were not used so much as a simple writing system, but rather as magical signs to be used for charms.
III. In the Proto-Germanic period, the god Tiwaz was associated with war, victory, marriage and the diurnal sky.
IV. The knowledge of the meanings attributed to the runes of the Younger Futhark is derived from the three Rune poems.
- a. All the above
b. ii and iv
c. i, ii and iv
d. i and iii
- Q40. Which of the following cannot be reasonably inferred with regard to the beliefs of the Proto-Germanic people?
- a. Odin came upon the runes after going through a lot of torment.
b. The name of a rune was almost always the first sound of a God's name
c. The cosmological power represented by a rune was invoked by writing it.
d. Proto-German Gods were modeled on humans.
- Q41. ABCD is a 4 digit number which gives a remainder of 5 when divided by 6. Find the remainder when ABCD is divided by 3?
- a. 0 b. 1 c. 2 d. cannot be determined
- Q42. A 240 m long train passes a pole in 24 seconds. How long will it take to cross a platform 650 m long?
- a. 65 sec b. 59 sec c. 100 sec d. 89 sec
- Q43. A does half as much work as B but in $\frac{3}{4}$ th of the time. If they work together, they finish a work in 18 days. Find the time taken by B alone to finish the work?
- a. 30 days b. 24 and $\frac{3}{4}$ c. 33 days d. 99 days
- Q44. A man can swim 5 km/hr in still water. If the speed of current be 3 km/hr, find the time taken by him to swim to a place 16 km upstream and back?
- a. 8 hr b. 7.5 hr c. 6.66 hr d. 10 hr
- Q45. Find the rate of compound interest applicable per annum if Rs.12000 amounts to Rs.20736 in a time of 3 years?
- a. 10% b. 25% c. 12% d. 20%
- Q46. Find the HCF of $\frac{2}{3}$, $\frac{8}{9}$, $\frac{64}{81}$ and $\frac{10}{27}$?
- a. $\frac{2}{3}$ b. $\frac{2}{81}$ c. $\frac{160}{3}$ d. $\frac{160}{81}$
- Q47. In how many different ways can we arrange the alphabets of the word "ANNUM" such that the vowels are not together?
- a. 36 ways b. 24 ways c. 60 ways d. 40 ways
- Q48. If the number $385a7b$ is divisible by 44, find the value of $a+b$?
- a. 7 b. 18 c. 11 d. cannot be determined
- Q49. In what proportion should water be mixed with milk in order to gain 32% by selling the mixture at a markup of 10 %?
- a. 2:3 b. 1:5 c. 1:3 d. 3:5
- Q50. A person sold two products for Rs.400 each, the first one at 25% profit and the second one at 50% loss. Find the overall profit or loss %?
- a. loss of 30% b. loss of 28.56% c. loss of 25% d. none of these
- Q51. How many pairs of numbers exist such that the HCF is 5 and LCM is 75?
- a. 0 b. 1 c. 2 d. 3
- Q52. In a round robin chess tournament each player played against every other player exactly once. If the total number of matches played is 171, find the number of players in the tournament?

- a. 17 b. 20 c. 19 d. none of these
Q53. In how many ways can 7 boys and 4 girls be seated in a straight line such that no two girls are together?
- a. $4! \times 7!$ b. $4! \times 7P4$ c. $7! \times 8P4$ d. $8! \times 7P4$
Q54. The ratio of the MP and the CP is 5:3 while the ratio of the CP and SP is 4:5. What is the percentage discount being offered?
- a. 33.33% b. 20% c. 25% d. cannot be determined
Q55. Find the sum of the digits of the smallest 3 digit number which will give remainder of 5 when divided by both 6 and 7?
- a. 8 b. 7 c. 6 d. 5
Q56. Find the largest number that will divide 251, 283 and 299 to give the same remainder in each case?
- a. 8 b. 17 c. 15 d. 16
Q57. A Household was able to save 10% of its income in every month. Since the income increased by 20%, the expenses were also increased by 20%. By what % is the new savings more or less than the earlier savings?
- a. 10% b. 70% c. 34% d. 20%
Q58. A box contains 2 white balls, 3 black balls and 4 red balls. In how many ways can 3 balls be drawn from the box if at least one black ball is to be included?
- a. 48 b. 64 c. 69 d. 71
Q59. A shopkeeper sells goods at a mark-up of 40%. Find the profit or loss % if his weighing balance shows 800 gms instead of 1 kg?
- a. 20% profit b. 75% profit c. 5% profit d. 12% profit
Q60. The selling price of 5 items is same as the cost price of 6 items. If the percentage mark-up is 10%, find the overall profit percentage?
- a. 20% b. 30% c. 32% d. 45%
Q61. A pipe can fill a tank alone in 15 hours but it takes 3 hours more due to a leak. In how much time can the leak alone empty half the tank?
- a. 90 hrs b. 60 hrs c. 75 hrs d. 30 hrs
Q62. A person wanted to complete a work in 40 days and employed 100 men for the job. After 20 days only 40% of the work was completed and so he employed 75 more women to assist the men. If a man does as much work as 3 women, find the total time required to complete the job?
- a. 40 days b. 44 days c. 24 days d. 20 days
Q63. What is the value of the number which when diminished by 5 is completely divisible by 15, 18, 21 and 24?
- a. 2525 b. 2520 c. 1715 d. none of these
Q64. The sales of a company increased by 5% in the month of March 2017 and again by y% in the month of April 2017. As a result, the sales in May 2017 is 36.5% more than the sales in February 2017. Find the value of y?
- a. 25% b. 31.5% c. 30% d. 29%
Q65. In how many ways can a group of 5 students be selected out of a group of 5 boys and 4 girls such that there is at least one girl in the group?
- a. 126 ways b. 121 ways c. 125 ways d. none of these
Q66. A certain sum of money becomes 3 times in 4 years at SI. In how many years will it become 15 times its initial value.
- a. 20 years b. 24 years c. 18 years d. 28 years
Q67. If 12 Men or 15 Women can do a job in 20 days working 6 hours per day, how many men are required to work with 5 women to complete a work in 9 days working for 5 hours per day?
- a. 12 Men b. 28 Men c. 26 Men d. 24 Men
Q68. 2 people A and B start from the opposite ends of a corridor. Their speeds are in the ratio 5:3 respectively. When they meet for the first time, B has covered 45 m. If B reaches the other end 10 seconds after meeting A, find the time taken by A to reach the other end after the meeting?
- a. 3 sec b. 15 sec c. 3.6 sec d. 12 sec
Q69. A can finish a job in 20 days. He works with C and together they finished the job taking 40% lesser time than that required had A worked alone. In how many days will C alone complete the work?
- a. 40 days b. 30 days c. 60 days d. 18 days

Q70. The price of a commodity increased by 9.09%. By what % should the consumption decrease so that the expenditure is same as earlier?

- a. 9.09% b. 10% c. 8.33% d. cannot be determined

Q71. A train normally takes 15 seconds for a 2000 m journey. On a particular day, after covering $\frac{3}{4}$ th of the distance, it's speed reduced to half of it's normal speed. How much more time will the train take today for the journey?

- a. 3.75 sec b. 7.5 seconds c. 30 seconds d. 22.5 seconds

Q72. A person purchased at the rate of 7 items for Rs.60 and sold at the rate of 6 items for Rs.70. Find the profit or loss in percentage terms?

- a. 34.23% profit b. 35.34% profit c. 36.11% profit d. cannot be determined

Q73. A pipe can fill a tank in 18 hours but takes 2 hours more because of a leak in the tank. If water flows out of the leak at the rate of 1 litres in every minute, find the volume of the tank?

- a. 1080 litres b. 3060 litres c. 10800 litres d. 5040 litres

Q74. The difference between the CI and the SI on a certain sum at 7% per annum for 2 years is Rs.2940. Find the principal?

- a. Rs.800000 b. Rs.300000 c. Rs.600000 d. Rs.450000

Q75. The CI for 2 years is Rs.607.50/- while the SI for 5 years is Rs.1500. Find the rate of interest?

- a. 5% b. 7.5% c. 2.5% d. 9%

Q76. Two trains start from A and B and move towards B and A respectively. Train from A starts at 1 pm and reaches B at 5 pm while the train from B starts at 12 noon to reach A at 3 pm. After how much time from 12 noon will the 2 trains meet?

- a. 2 and $\frac{1}{7}$ hours b. 1 and $\frac{1}{7}$ hours c. 2 and $\frac{2}{7}$ hours d. 1 and $\frac{2}{7}$ hours

Q77. A person purchased 720 grams of a commodity for Rs.40. If the person wishes to make a profit of 20%, what is the quantity to be sold for Rs.40?

- a. 700 grams b. 650 grams c. 620 grams d. 600 grams

Q78. A pipe can fill a tank in 28 hours while a leak can empty the tank in 14 hours. Initially, the tank is full and the pipe is closed when the leak is opened. After 5 hours, the pipe is also opened. In how much time will the tank get empty?

- a. 18 hours b. 23 hours c. 10 hours d. none of these

Q79. When the CP of an article is decreased by Rs.60/- keeping the selling price same, the profit % changes from 10% to 25%. Find the cost price?

- a. Rs.150 b. Rs.300 c. Rs.350 d. Rs.400

Q80. An increase of 25% in the price of onions results in a 5 Kg decrease in the quantity of Onions that can be purchased for Rs.300. Find the original price of Onions per Kg before the increase?

- a. Rs.12/Kg b. Rs.15/Kg c. Rs.20/Kg d. Rs.25/Kg

Q81. If the amount after 5 years is Rs.3200 and the amount after 8 years is Rs.3800 both at simple interest, find the rate of interest applicable?

- a. cannot be determined b. 9.09% c. 8.33% d. 10%

Q82. A can do a piece of work in 36 days B in 54 days and C in 72 days. All three began the work together but A left 8 days and B 12 days before the completion of the work. For how many days in all did C work?

- a. 24 days b. 28 days c. 32 days d. 36 days

Q83. A man works twice as fast as a woman. A woman works twice as fast as a child. If 16 men can complete a job in 12 days, how many days would be required for 32 women and 64 boys together to complete the same job?

- a. 2 day b. 3 days c. 4 days d. 6 days

Q84. A man on a motorcycle is moving on a straight road at the speed of 54 kmph. Half a minute later a car starts from the same point and moves in the same direction at a speed of 20 m/sec. How far from the start will the man on the motorcycle reach before he is overtaken by the car?

- a. 1800 m b. 1500 m c. 1200 m d. 1000 m

Q85. The speed of a boat in still water is 5 kmph while the speed of the stream is x kmph. A person on the boat can see a log of wood at a distance of 12 km floating towards the boat. In how much time will the boat reach the log of wood?

- a. 2 hrs 40 minutes b. 2 hrs 24 minutes c. cannot be determined d. 3 hrs

Q86. When the selling price of an item is doubled keeping the cost price at the same level, the value of the profit generated triples. Find the original profit percentage?

- a. 50% b. 75% c. 100% d. cannot be determined
- Q87. A person was travelling at a speed of 50 kmph. If he reduced the speed by 10%, it would take him 10 minutes more to reach his destination. What is the usual time taken by the person for the journey?
- a. 1 hour b. 1.5 hour c. 2 hours d. 2.5 hours
- Q88. A total amount of Rs.1200 was invested at simple interest, one part at 4% interest per annum and the remaining at 10% interest per annum. If the overall return at the end of one year was 8%, find the amount invested at 4% per annum?
- a. Rs.800 b. Rs.600 c. Rs.400 d. Rs.200
- Q89. A work can be completed by 16 men and 8 women in 48 days. How many days will be required if the number of men decrease to 12 and the number of women decrease by 2?
- a. 36 days b. 64 days c. 60 days d. 70 days
- Q90. A boy rides to his school on a cycle and walks back home taking a total time of 4 hours. He would take 2 hours more if he decides to walk both ways. Find the total time taken if he decides to ride both ways?
- a. 1.5 hours b. 3 hours c. 2 hours d. 2.5 hours
- Q91. A can complete a work in 35 days and B can complete the same work in 20 days. They started to work together but A left 3 days before the work was completed. Find the total number of days for which B worked?
- a. 10 and 9/11 days b. 11 and 3/11 days c. 13 and 9/11 days d. none of these
- Q92. In a society, the number of men and women is in the ratio 5:3. In a competition, 60% of the residents in the society participated. If 50% of the women participated, find the percentage of men who participated in the competition?
- a. 66% b. 33% c. 15% d. 22%
- Q93. The total interest received on a particular principal at the end of 8 years is equal to 80% of the sum invested. At the same rate of interest, find the compound interest on a sum of Rs.2000 for 2 years?
- a. Rs.360 b. Rs.500 c. Rs.420 d. Rs.320
- Q94. A boat takes 90 minutes less to travel 36 km downstream than to travel the same distance upstream. If the speed of the boat in still water is 10 kmph, find the speed of the stream?
- a. 2 kmph b. 2.5 kmph c. 3 kmph d. 4 kmph
- Q95. A man invested Rs.4000 at a certain rate of interest compounded per annum. After 2 years, the sum of money including interest was Rs.12000. After how many years from the start will the sum including interest become Rs.324000?
- a. 8 years b. 9 years c. 6 years d. none of these
- Q96. The side of a square was increased by 3 cm and therefore the area of the square increased by 10.25%. Find the changed dimension of the side of the square?
- a. 30 cm b. 63 cm c. 60 cm d. 33 cm
- Q97. The sum of a 2 digit number and the number formed by reversing it's digits is exactly divisible by 132. If the unit's place of the number is 2 more than the ten's place, find the remainder when the number is divided by 5?
- a. 2 b. 0 c. 1 d. 3
- Q98. Find the unit's place of the number 3498^{A9843}
- a. 8 b. 4 c. 2 d. 6
- Q99. A man is seated in a 150 m long train. Another train coming from the opposite direction at 25 m/sec crosses the man in 5 seconds. Find the time taken by the second train to cross a bridge 175 m long?
- a. 12 sec b. 8 sec c. 10 sec d. none of these
- Q100. If 9 men can make 9 bags in 9 days, find the number of bags that 27 men can make in 27 days?
- a. 27 b. 9 c. 36 d. 81